

# PROJECT REPORT

## Exploratory Data Analysis (EDA)

DecodeLabs Data Analytics Internship  
Industrial Training Kit | Batch 2026

|               |                                                     |
|---------------|-----------------------------------------------------|
| Project Title | Exploratory Data Analysis (EDA)                     |
| Organization  | DecodeLabs                                          |
| Program       | Data Analytics Internship – Industrial Training Kit |
| Batch         | 2026                                                |
| Tool Used     | Python + Pandas + Matplotlib/Seaborn                |
| Date          | 4 September 2026                                    |

### 1. Introduction

This report presents the Exploratory Data Analysis (EDA) performed on the cleaned e-commerce order dataset. The objective was to move beyond static tables and uncover patterns, trends, distributions, and outliers that translate into actionable business insights.

EDA is the diagnostic phase of analytics. Before dashboards or predictive models are built, the analyst must understand the shape of the data, locate its center of gravity, identify noise versus signal, and answer the critical “So What?” question for stakeholders.

### 2. Project Objectives

- Calculate basic statistics (mean, median, count, distribution).
- Identify trends and outliers using IQR and visual methods.
- Explore relationships through correlation analysis.
- Summarize key observations with clear business impact.
- Produce reproducible code and stakeholder-ready visualizations.

### 3. Dataset Overview

The analysis uses the cleaned order dataset produced in the previous data-cleaning milestone. Summary profile:

| Attribute        | Value                                                    |
|------------------|----------------------------------------------------------|
| Total Orders     | 1,200                                                    |
| Date Range       | 2023-01-01 to 2025-06-30                                 |
| Unique Customers | 1,189                                                    |
| Unique Products  | 7 (Phone, Laptop, Tablet, Monitor, Printer, Chair, Desk) |
| Total Revenue    | \$1,264,761.96                                           |

### 4. Methodology

#### 4.1 Descriptive Statistics

Univariate analysis was performed on Quantity, UnitPrice, ItemsInCart, and TotalPrice. Mean, median, standard deviation, and the five-number summary (min, Q1, median, Q3, max) were calculated to understand central tendency and spread.

#### 4.2 Distribution & Skew

TotalPrice is right-skewed (mean \$1,054 > median \$824). This indicates a minority of high-value orders pulling the average upward — a classic pattern in e-commerce order data.

#### 4.3 Outlier Detection (IQR Method)

The IQR rule ( $\text{Outlier} < Q1 - 1.5 \times \text{IQR}$  or  $> Q3 + 1.5 \times \text{IQR}$ ) was applied to TotalPrice. Only 8 orders (0.7%) were flagged as outliers. These represent high-value transactions and should be treated as potential signal (VIP / bulk orders) rather than noise.

#### 4.4 Correlation Analysis

Pearson correlation was computed among numeric fields. The strongest relationships are UnitPrice ↔ TotalPrice (r = 0.72) and Quantity ↔ TotalPrice (r = 0.62). ItemsInCart and Quantity are moderately correlated (r = 0.65). Correlation is a clue, not causation — no causal claims are made.

## 5. Key Findings

| Metric                | Value      | Interpretation                       |
|-----------------------|------------|--------------------------------------|
| Avg Order Value       | \$1,053.97 | Pulled up by high-value orders       |
| Median Order Value    | \$823.62   | More representative of typical order |
| Cancellation Rate     | 20.8%      | High — investigate friction points   |
| Return Rate           | 20.6%      | High — product/expectation mismatch? |
| Outliers (TotalPrice) | 8 (0.7%)   | Likely signal (high-value orders)    |

## 6. Business Diagnosis (“So What?”)

**What the data says:** Cancellation and return rates together exceed 41%. Average order value is inflated by a right-skewed distribution.

**Business impact:** Nearly 1 in 5 orders is cancelled and another 1 in 5 is returned. This represents significant lost revenue and operational cost. Product mix is relatively balanced; no single product dominates revenue.

**Recommended next steps:** Investigate root causes of cancellations and returns (delivery time, product description accuracy, payment friction). Segment high-value outlier customers for retention offers. Use median (not mean) when setting order-value targets.

## 7. Deliverables

- cleaned\_dataset.xlsx – Input dataset used for analysis
- eda\_analysis.py – Fully reproducible EDA script
- eda\_outputs/ – Charts, correlation matrix, outlier list, summary tables
- Project\_Report.docx – This formal project report
- README.md – Project documentation for GitHub

## 8. Conclusion

This EDA successfully transformed a cleaned transactional dataset into a clear diagnostic picture. Distributions, outliers, correlations, and business rates were quantified and translated into concrete observations. The analysis confirms that data integrity from the previous milestone enabled reliable exploration, and highlights high cancellation/return rates as the strongest operational signal for further investigation.

EDA is the bridge between clean data and decision-ready insight. This project demonstrates that bridge in practice.